



Project 4

Quantum Error

Mitigation

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Zero-Noise Extrapolation

Noise-Scaling

Unitary folding

$$U \rightarrow U(U^\dagger U)^n$$

- Circuit folding

$$U \rightarrow U(U^\dagger U)^n L_d^\dagger L_{(d-1)}^\dagger \dots L_s^\dagger L_s \dots L_{(d-1)} L_d$$

- Layer (gate) folding

$$L_j \rightarrow \begin{cases} L_j (L_j^\dagger L_j)^n & \text{if } j \notin S, \\ L_j (L_j^\dagger L_j)^{n+1} & \text{if } j \in S. \end{cases}$$

Parameter Noise Scaling

- Parametrized quantum gate

$$G(\theta) = \exp\left(-i \sum_{j=1}^l \theta_j H_j\right)$$

- Parameter (finite precision)

$$\theta_j' = \theta_j + \hat{\epsilon}_j$$

- Parameter (additional classical noise)

$$\begin{aligned} \theta_j' &\rightarrow \theta_j + \hat{\delta}_j \\ \theta_j' &= \theta_j + \sqrt{\lambda} \hat{\epsilon}_j \end{aligned}$$

Extrapolation: Non-Adaptive

Polynomial Extrapolation

$$E_{poly}^{(d)}(\lambda) = c_0 + c_1\lambda + \dots + c_d\lambda^d$$

- Linear extrapolation

$$E_{linear} = E_{poly}^{(d=1)}(\lambda) = c_0 + c_1\lambda$$

- Richardson extrapolation

$$E_{Rich} = E_{poly}^{(d=m-1)}(\lambda) = c_0 + c_1\lambda + \dots + c_{m-1}\lambda^{m-1}$$

Poly-Exponential Extrapolation

$$E_{polyexp}^{(d)}(\lambda) = a \pm e^{z(\lambda)},$$
$$z(\lambda) := z_0 + z_1\lambda + \dots + z_d\lambda^d$$

- Exponential Extrapolation

$$E_{exp}(\lambda) = E_{polyexp}^{(d=1)}(\lambda) = a + be^{-c\lambda}$$

Extrapolation: Adaptive

- Exponential ansatz

- Estimators

- $\hat{b} = (y_1 - a)^{\frac{\lambda_2}{\lambda_2 - \lambda_1}} (y_2 - a)^{-\frac{\lambda_1}{\lambda_2 - \lambda_1}}$

- $\hat{c} = \frac{1}{\lambda_2 - \lambda_1} \log \frac{y_1 - a}{y_2 - a}$

- $E(\lambda_1)$ and $E(\lambda_2)$ with variance $\sigma_1^2 = \frac{\sigma_0^2}{N_1}$ and $\sigma_2^2 = \frac{\sigma_0^2}{N_2}$

- $MSE(\hat{b}) = \left(\frac{\partial \hat{b}}{\partial y_1}\right)^2 \sigma_1^2 + \left(\frac{\partial \hat{b}}{\partial y_2}\right)^2 \sigma_2^2 = \frac{\sigma_0^2}{(\lambda_2 - \lambda_1)^2} \left[\frac{\lambda_2^2 e^{2c\lambda_1}}{N_1} + \frac{\lambda_1^2 e^{2c\lambda_2}}{N_2} \right]$

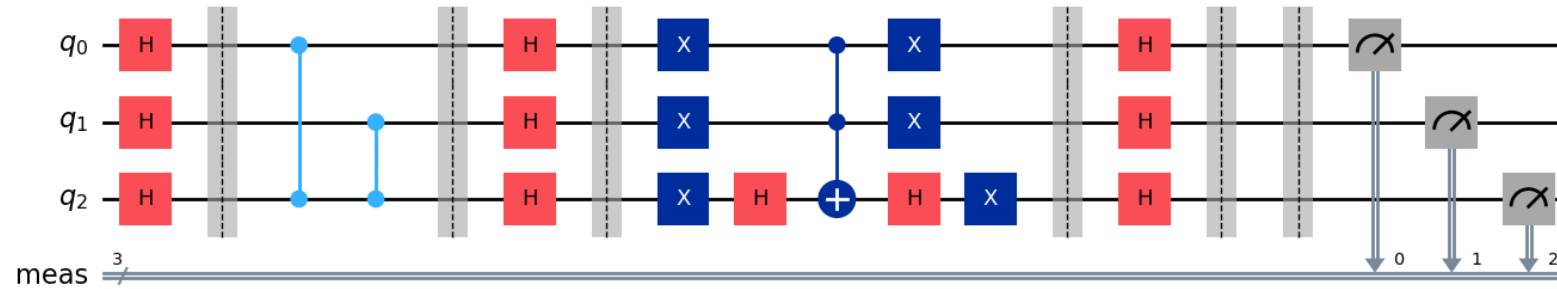
Extrapolation: Adaptive

- Error minimization
- $N_1 = N_{\max} \frac{\lambda_1}{\lambda_1 + \lambda_2 e^{-c(\lambda_2 - \lambda_1)}}$
- $N_2 = N_{\max} \frac{\lambda_2 e^{-c(\lambda_2 - \lambda_1)}}{\lambda_1 + \lambda_2 e^{-c(\lambda_2 - \lambda_1)}}$
- $MSE(\hat{b}) = \sigma_0^2 \left[\frac{\lambda_2 e^{c\lambda_1} + \lambda_1 e^{c\lambda_2}}{\lambda_2 - \lambda_1} \right]^2$
- $e^{c(\lambda_2 - \lambda_1)}(c(\lambda_2 - \lambda_1) - 1) - 1 = 0$
- $c(\lambda_2 - \lambda_1) = \alpha, \quad \alpha \simeq 1.27846$

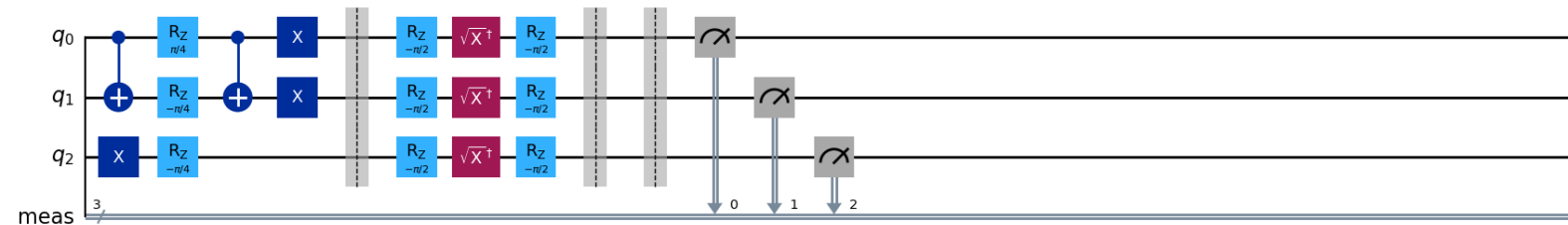
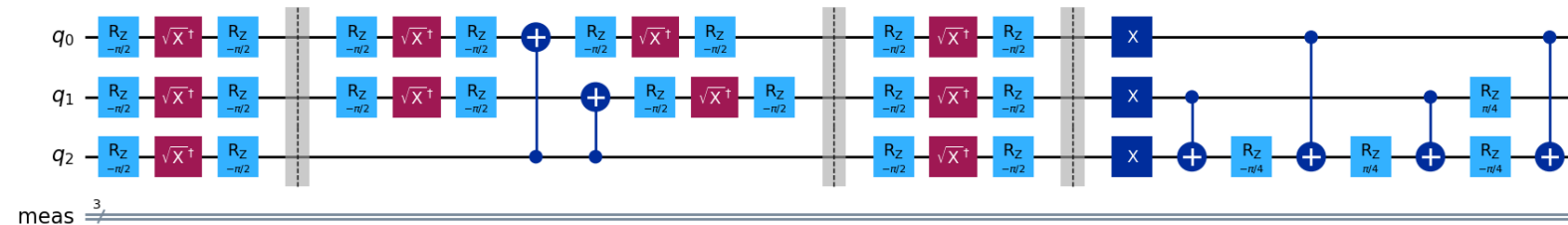


Simulation

Grover's Circuit – Transpiled

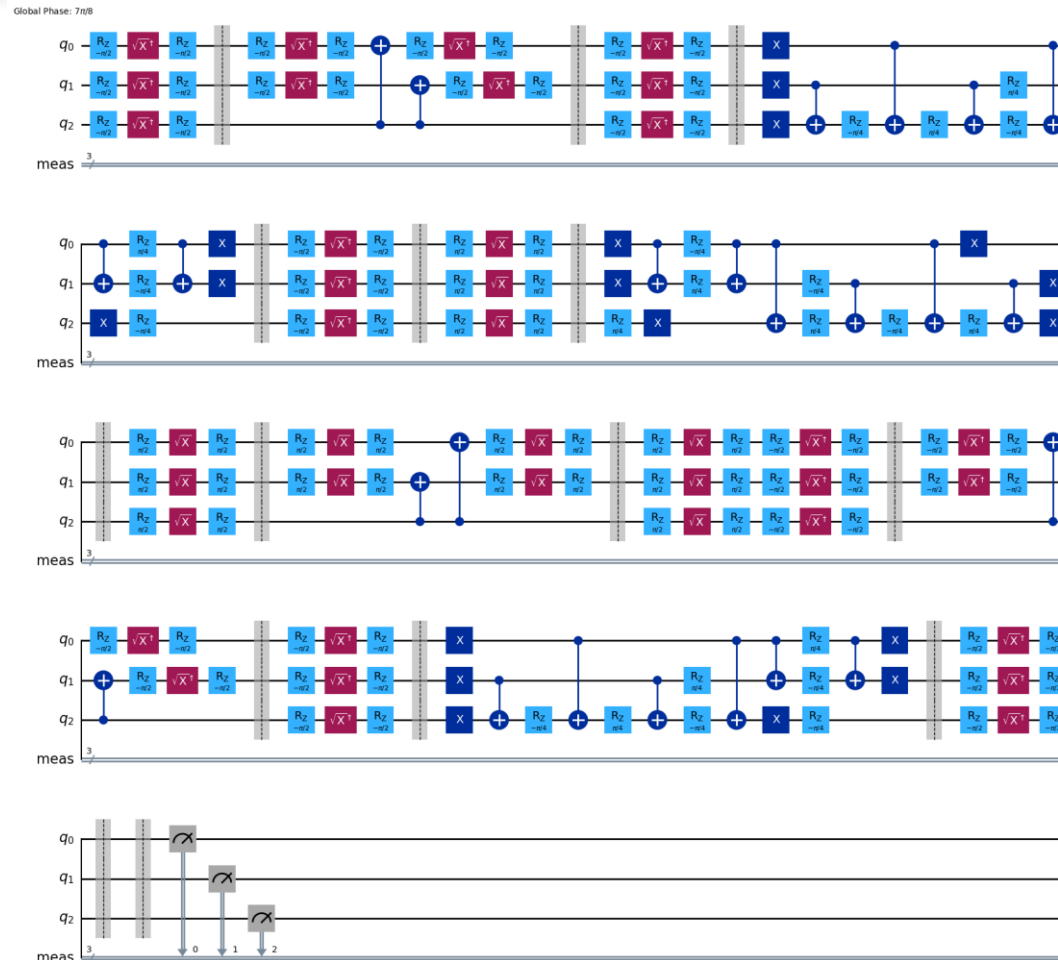


Global Phase: $7\pi/8$



Grover's Circuit – Circuit Folding

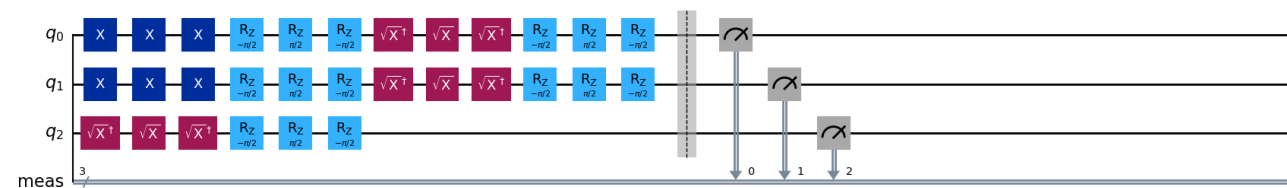
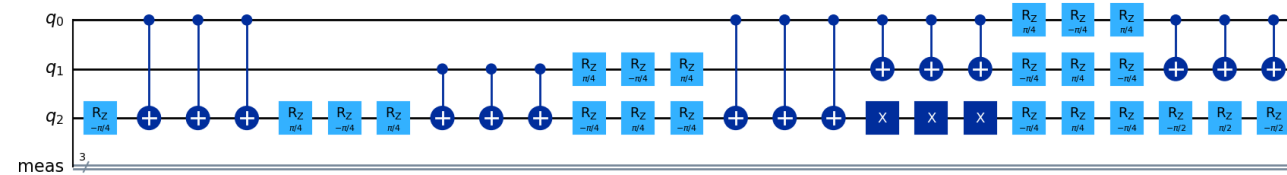
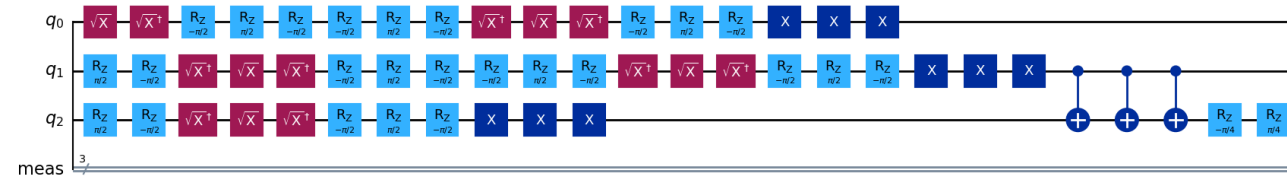
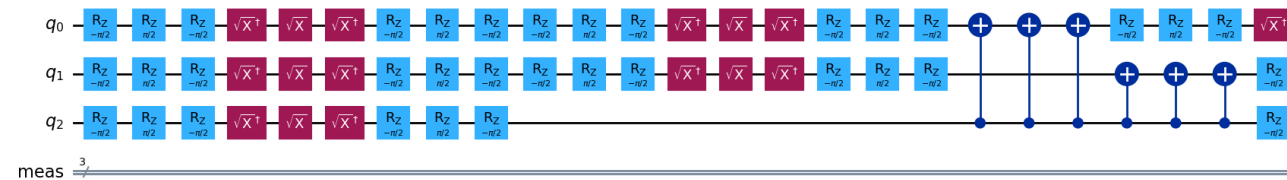
$(\lambda = 3)$



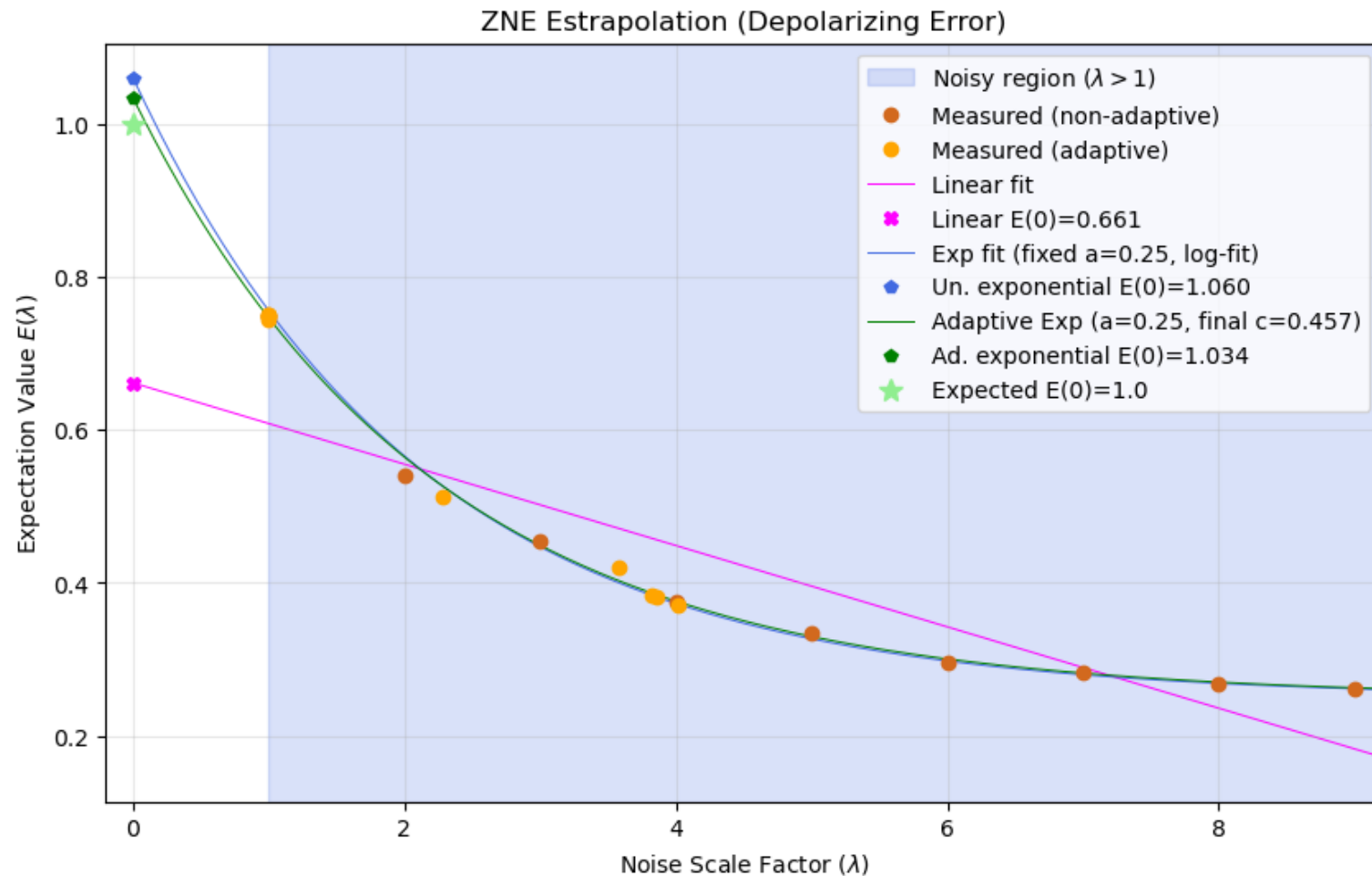
Grover's Circuit – Gate Folding

$(\lambda = 3)$

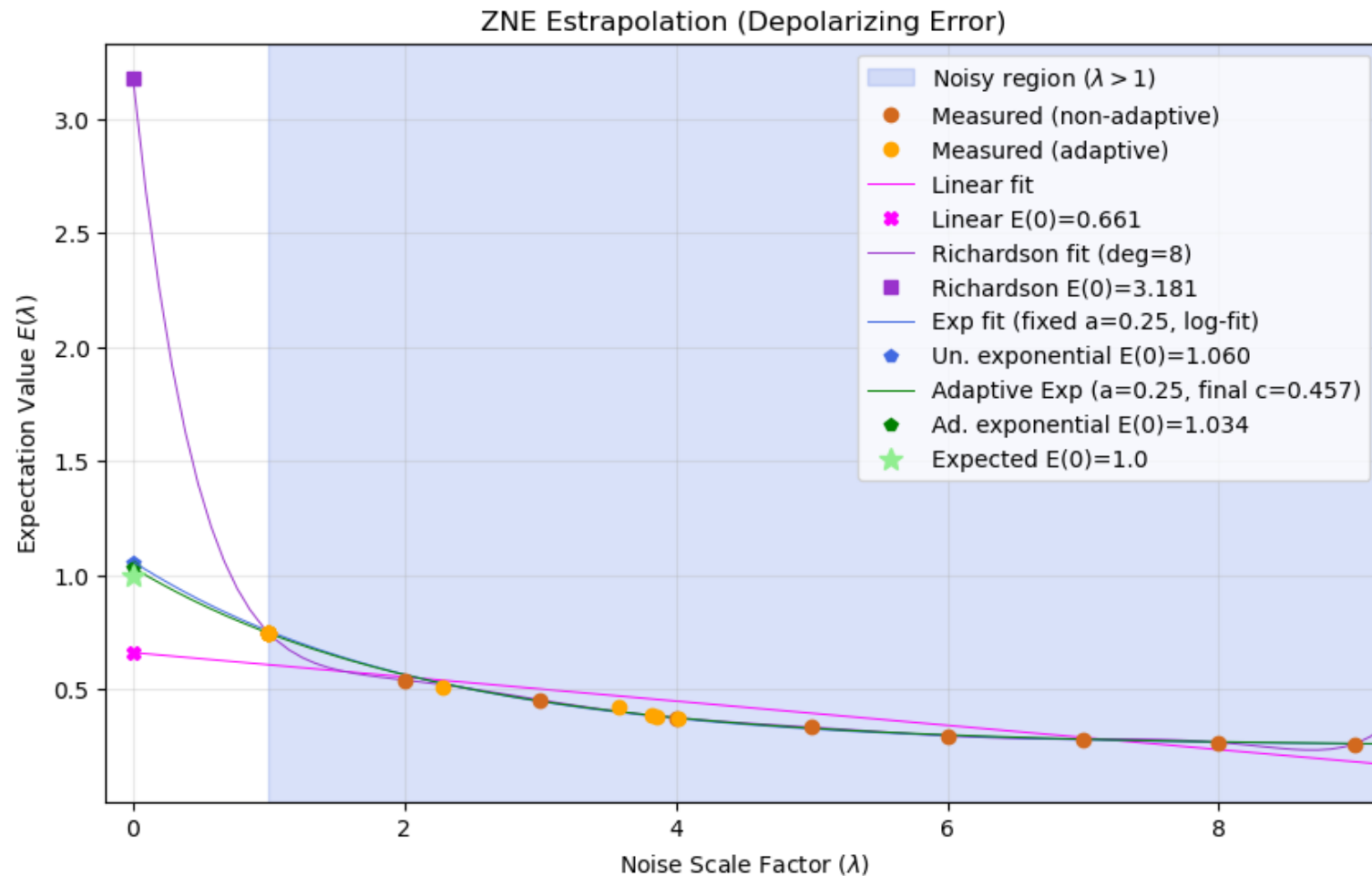
Global Phase: $7\pi/8$



Grover's Circuit – ZNE



Grover's Circuit – ZNE



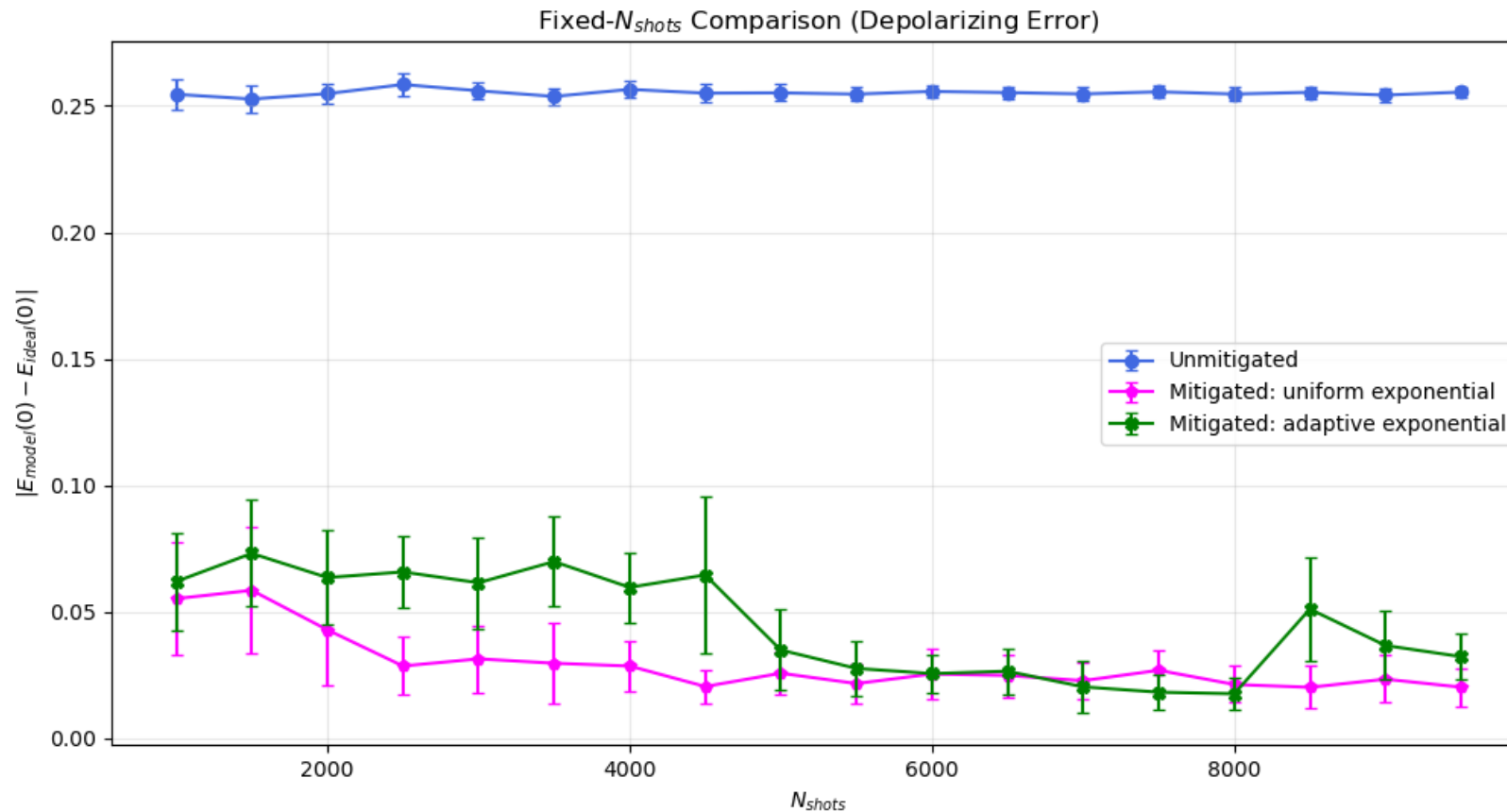


Quantitative Analysis

ZNE – Overview

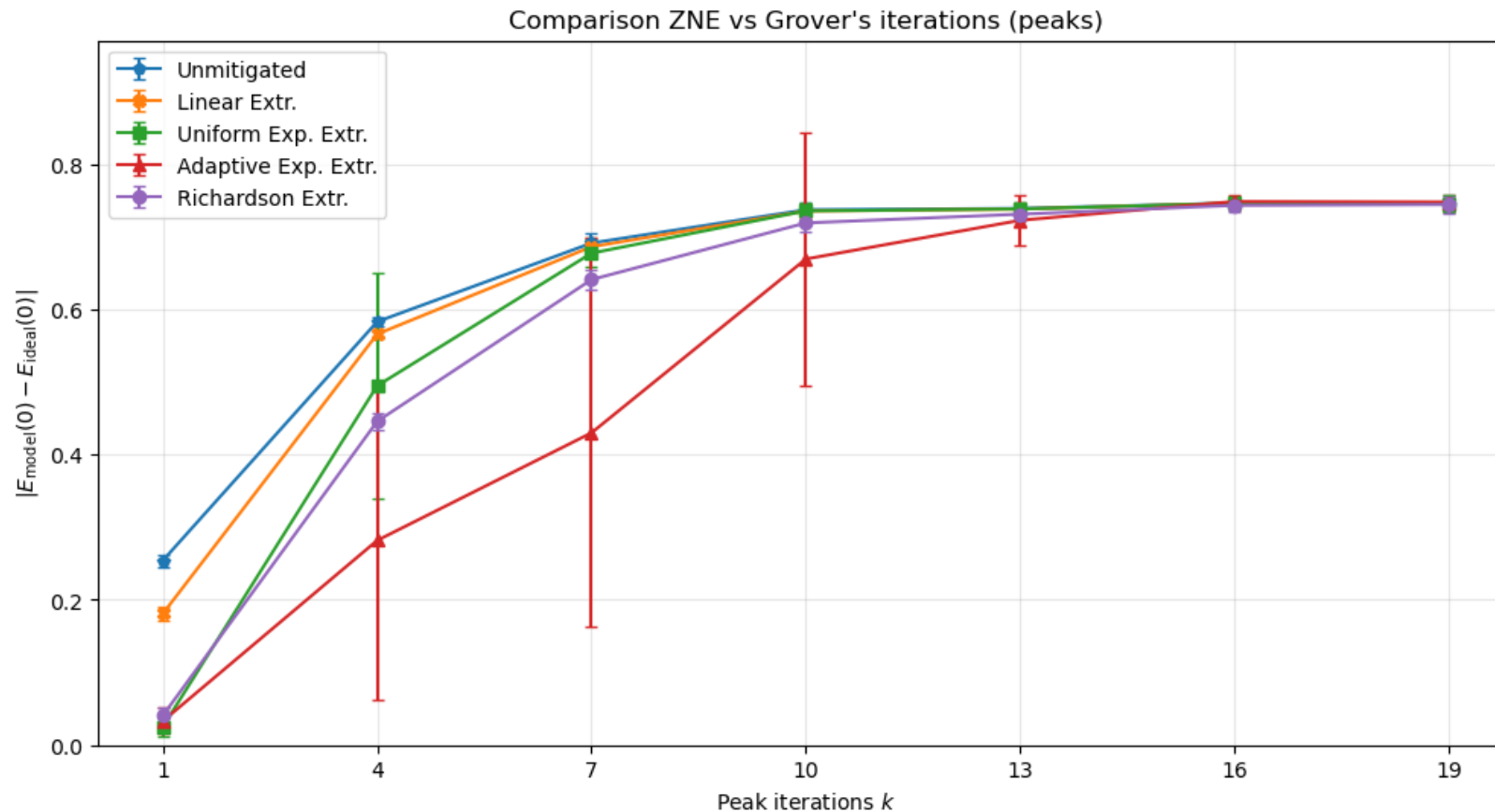
- Gate-level access to the system
- High circuit complexity (coherent time of the hardware)
- Unitary folding doesn't necessarily correspond to an effective scaling of the noise
- Noise model varies effectiveness
 - Depolarizing errors
 - Thermal Errors
 - Coherent Unitary errors
 - SPAM
- Non-physical extrapolation

Comparison Adaptive vs Non-Adaptive Fixed Shots

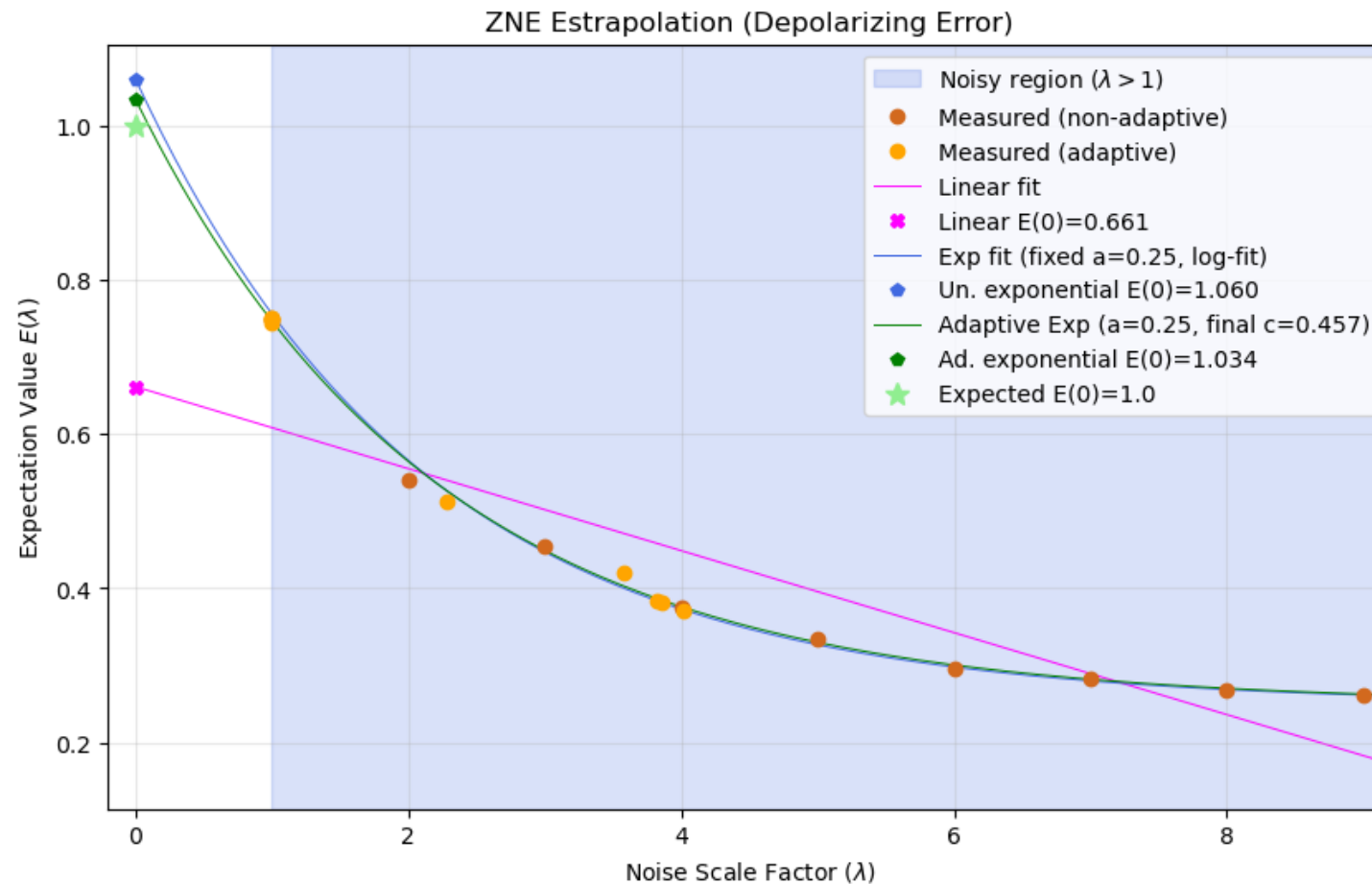


Comparison ZNE techniques

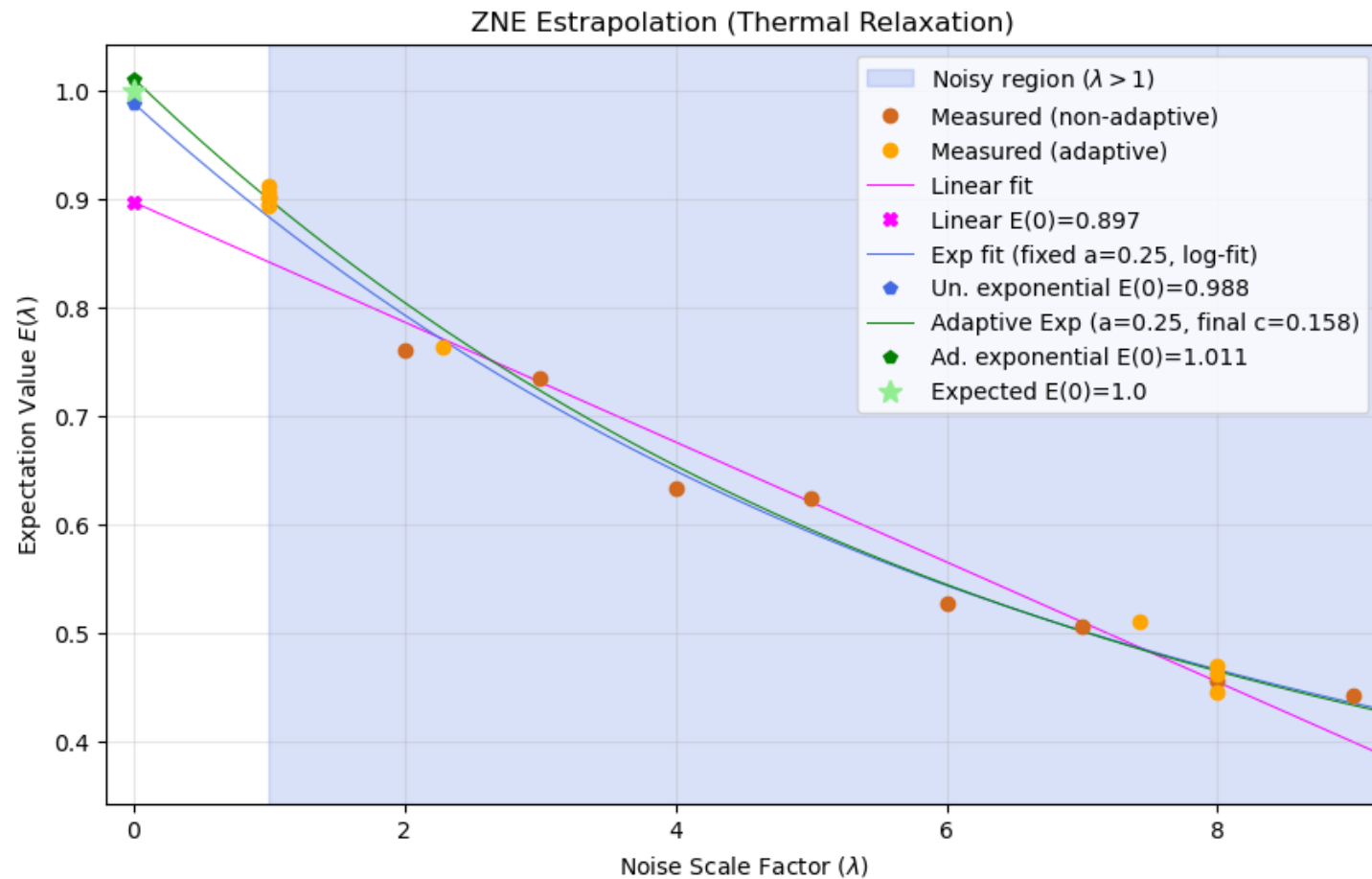
Varying Depth



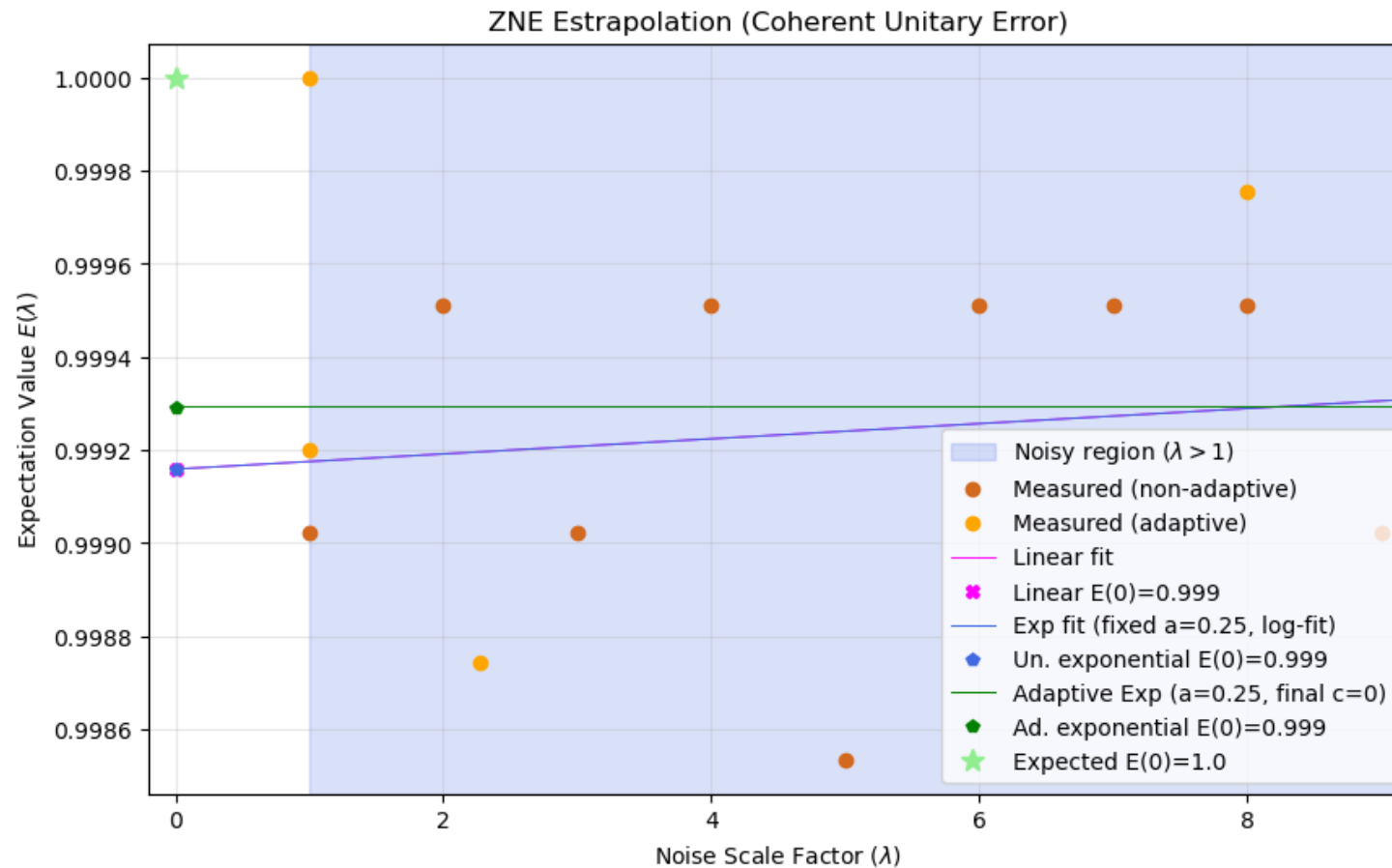
Depolarizing Noise - Exponential Ansatz



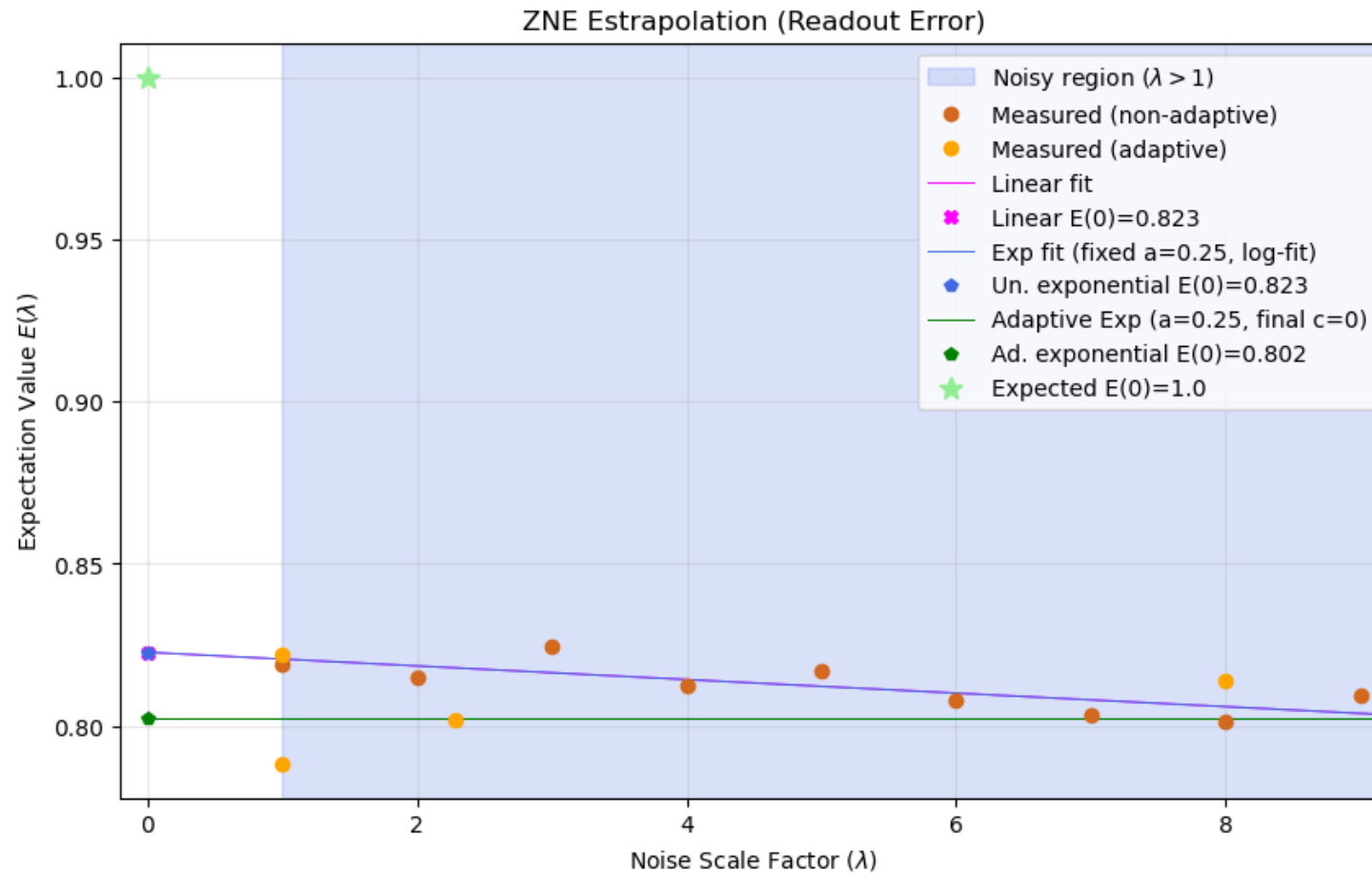
Thermal Relaxation



Coherent Errors



Readout Errors





arXiv:2005.10921 [quant-ph]
